Machine Learning

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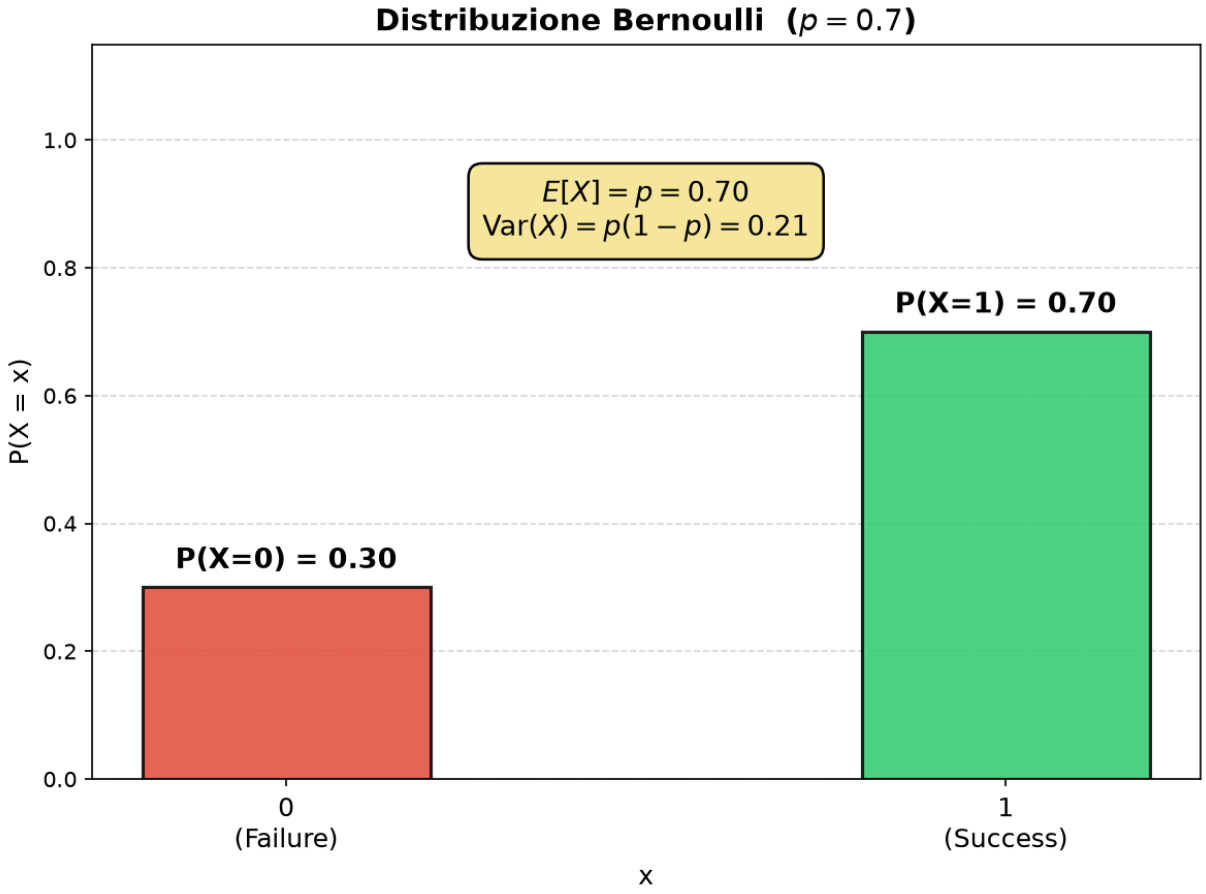
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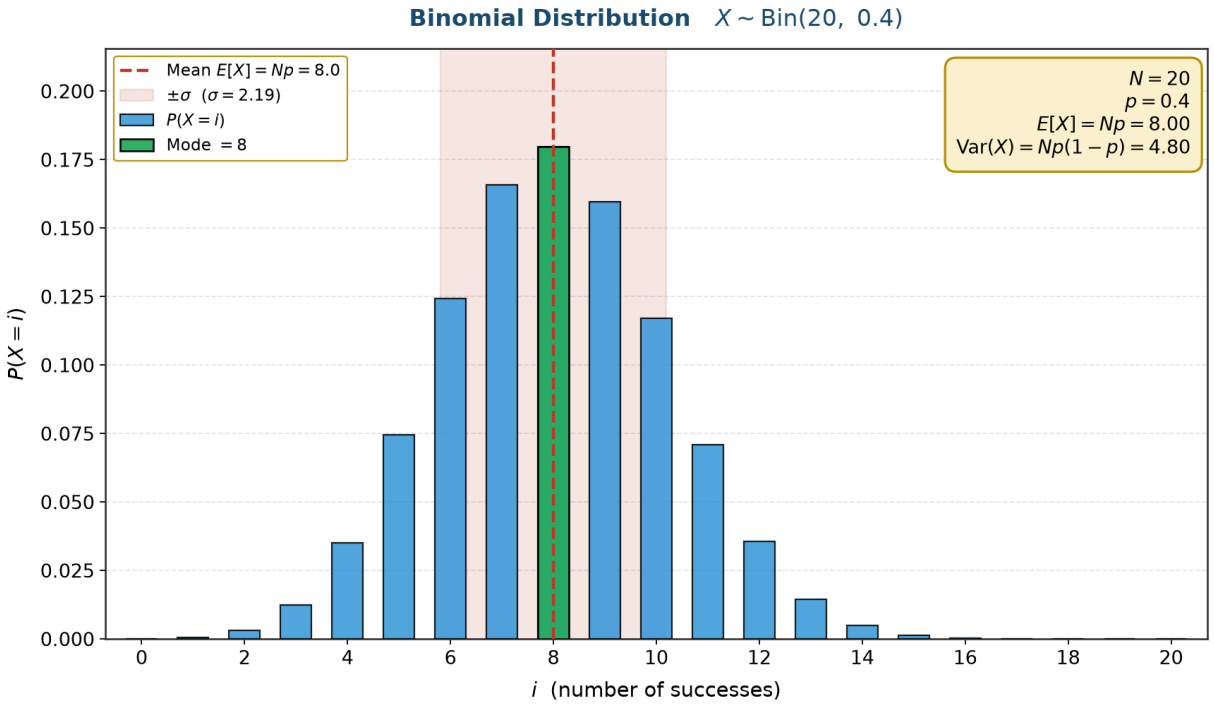
1. Math for Machine Learning

1.1. Probability

- **Random Experiment**: It's and experiment whose outcome is not predictable with certainty in advance.
- **Probability**: When an experiment is repeated under the same conditions, for any event E , the proportion of time that the outcome is in E approaches some costant value which is the probability of the event noted as $P(E)$. You could also say that the probability is a *degree of belief* in the occurence of the event E .
- **Axioms of Probability**: a series of statements that are certainly true in the world of probability.
  - $0 \leq P(E) \leq 1$.
    - E_1 certain $\iff P(E_1) = 1$.
    - E_2 never happening $\iff P(E_2) = 0$.
  - Given S a sample space of all possible outcomes, $P(S) = 1$.
  - If $E_i, i = 1, ..., n$ are mutually exclusive we have $P\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} P(E_i)$.
  - $P(E^c) = 1 - P(E)$, where E^c denotes the complement of E.
  - $P(E \cup F) = P(E) + P(F) - P(E \cap F)$ given $E \cap F \neq \emptyset$.
- **Conditional Probability**: is the likelihood of an event E happening based on the knowledge that another related event F has already occurred.
  - $P(E|F)$ is the probability of the occurrence of event E given that F occurred and is given as $P(E|F) = \frac{P(E \cap F)}{P(F)}$.
  - **Bayes Formula**: $P(F|E) = \frac{P(E|F)P(F)}{P(E)}$.
  - E and F are independent when $P(E|F) = P(E)$, also $P(E \cap F) = P(E)P(F)$.
- **Mean**: The mean of a random variable X is the average value of X in a large number of experiments defined as $E[X] = \sum_i x_i P(x_i)$ and given $a, b \in \mathbb{R}$ we have the following properties:
  - $E[aX + b] = aE[X] + b$.
  - $E[X + Y] = E[X] + E[Y]$.
  - $E[g(X)] = \sum_i g(x_i)P(x_i)$.
  - $E[X^n] = \sum_i x_i^n P(X_i)$, which is the *nth moment*.
- **Variance**: The variance measures how much X varies around the expected value. If $\mu = E[X]$, the vartiance is defined with the following formula $\mathrm{Var}(X) = E[(X - \mu)^2] = E[X^2] - \mu^2$. Also, usually for variance we use the symbol σ^2 , $\sigma(X) = \sqrt{\mathrm{Var}(X)}$ is known as the **standard deviation**.
- **Distributions**: we consider both discrete cases and continuous cases.
  - **Discrete Case**:
    - **Bernoulli**: Given that output 1 is success 0 is failure and p is the probability of success. We have:
      - $P(X = 1) = p$.
      - $P(X = 0) = 1 - p$.
      - $E[X] = p$.
      - $\mathrm{Var}(X) = p(1 - p)$.

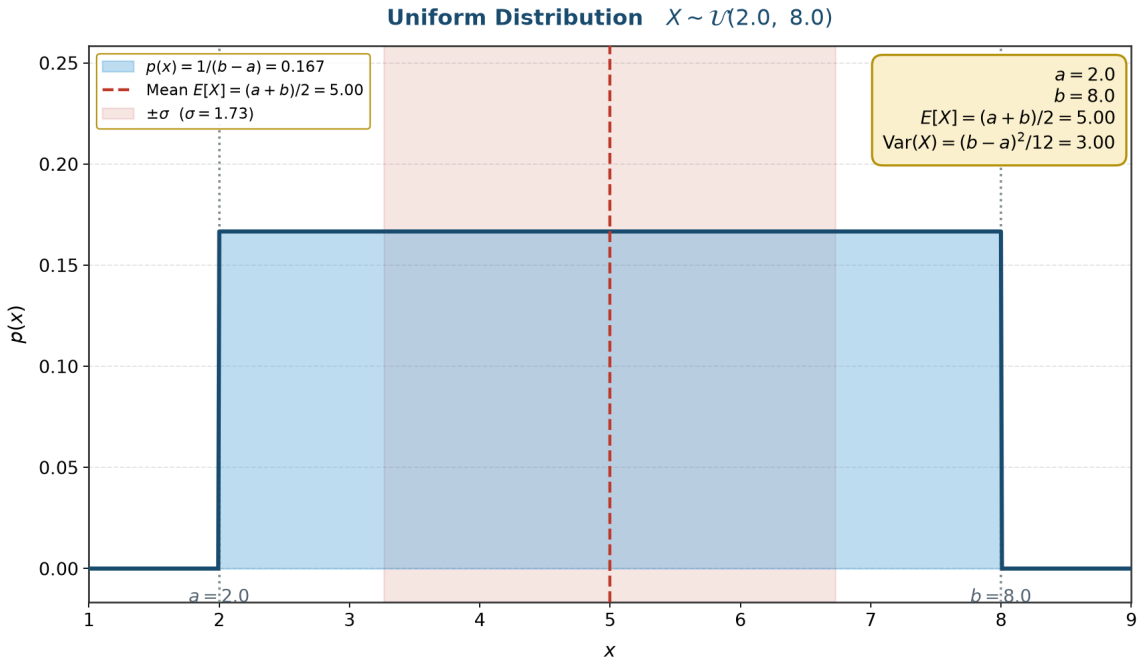


- **Binomial**: If N identical independent Bernoulli trials are made, the random variable X that represent the number of successes that occur in N trials is binomial distributed.
  - $P(X = i) = \binom{N}{i} p^i (1-p)^{N-i}, i = 1, ..., N$.
  - $E[X] = Np$.
  - $\mathrm{Var}(X) = Np(1-p)$.

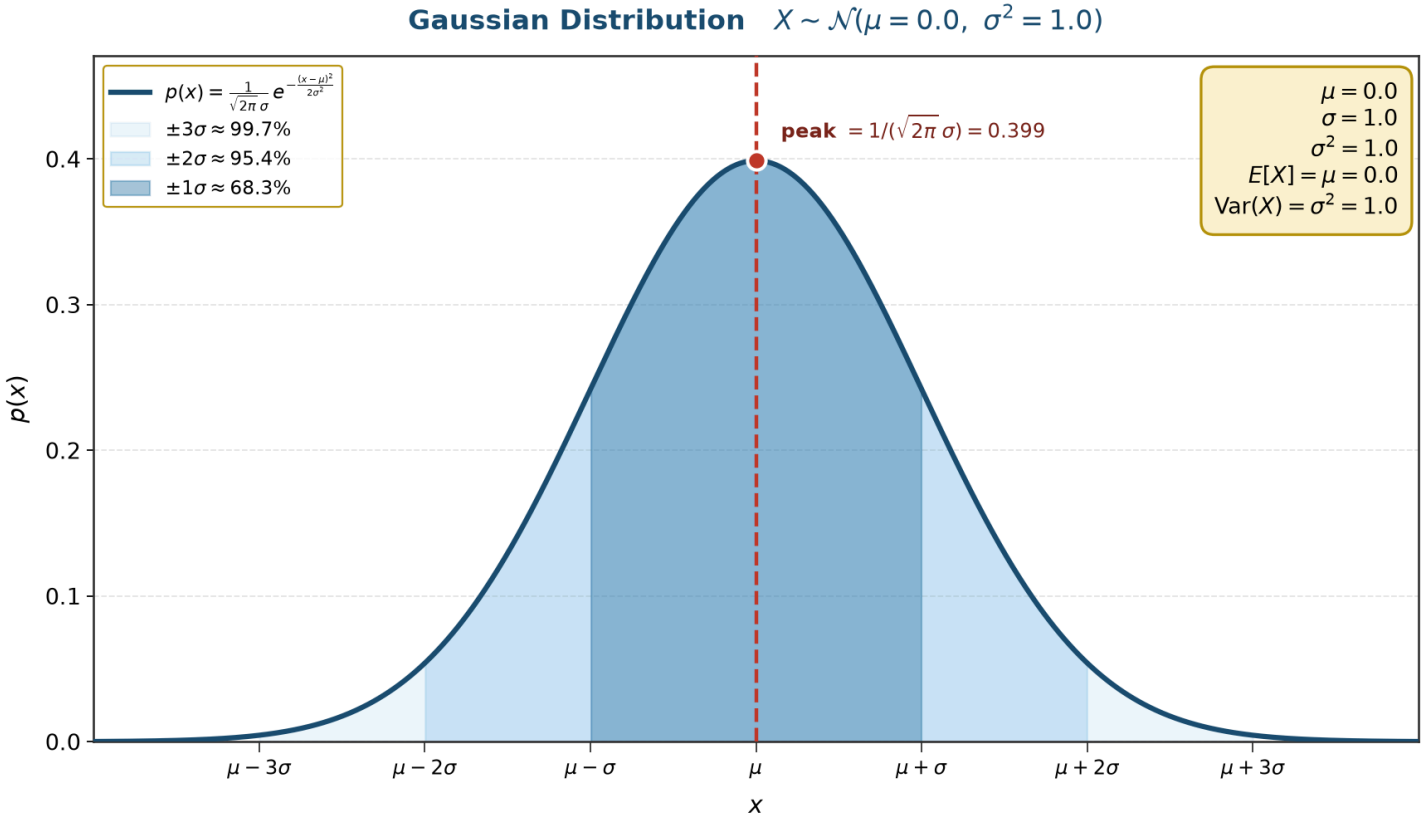


- **Continuous Case**:

- **Uniform**: In the interval $[a, b]$.
  - $p(x) = \frac{1}{b-a}$ if $a \leq x \leq b$ and $p(x) = 0$ otherwise.
  - $E[X] = \frac{a+b}{2}$.
  - $\text{Var}(X) = \frac{(b-a)^2}{12}$.



- **Gaussian**: Uses μ and σ^2 , $N(\mu, \sigma^2)$:
  - $p(x) = \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), -\infty < x < +\infty$



1.2. Vectors and Matrices

- **Vector**: A n-dimensional vector $x \in \mathbb{R}^n$ is a collection of n scalar values arranged in a column:

$$\begin{pmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{pmatrix}$$

  Given $x, z \in \mathbb{R}^n$ and $q \in \mathbb{R}$ we have the following operations:

  - **Sum**: $\begin{pmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{pmatrix} + \begin{pmatrix} z_1 \\ z_2 \\ \dots \\ z_n \end{pmatrix} = \begin{pmatrix} x_1 + z_1 \\ x_2 + z_2 \\ \dots \\ x_n + z_n \end{pmatrix}$.
  - **Multiplication by scalar**: $q \begin{pmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{pmatrix} = \begin{pmatrix} qx_1 \\ qx_2 \\ \dots \\ qx_n \end{pmatrix}$.
  - **Dot product**: $x \cdot z = \sum_i x_i z_i$.
    - **Geometrical Interpretation**: $x \cdot z = |x|\ |z| \cos(\theta)$ where θ is the angle between the two vectors.
  - **Length of a vector**: $|x| = \sqrt{\sum_i x_i^2}$.

- **Binary Vector**: Now we consider the case of a n-dimensional vector $x \in \{0,1\}^n$, we can use it to describe a set inside a universe of n elements where x is the set containing the elements corresponding to the ones in the vector.
  - **Cardinality of the set**: $|x|^2$
  - **Cardinality of the intersection**: $x \cdot z$ where $x, z \in \{0,1\}^n$
  - **Projection of x on z** : $x_z = \frac{(x \cdot z)}{|z|} \frac{z}{|z|} = \left(\frac{x \cdot z}{z \cdot z}\right) z = \alpha z$
    - $\alpha = P(x|z)$

- **Matrix**: A matrix $A \in \mathbb{R}^{m \times n}$ is a collection of scalar values arranged in a rectangle of m rows and n columns.

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

  - $a_{ij} = [A]_{ij}$
  - Given $x \in \mathbb{R}^n$, we can also say that $x \in \mathbb{R}^{n \times 1}$.
  - **Sum**: $[A + B]_{ij} = [A]_{ij} + [B]_{ij} = a_{ij} + b_{ij}$
  - **Product**: $[AB]_{ij} = \sum_{q=1}^{k} a_{iq} b_{qj}$ given $A \in \mathbb{R}^{m \times k}$ and $B \in \mathbb{R}^{k \times n}$.
    - Generally $AB \neq BA$
  - **Transpose**: The transpose of $A \in \mathbb{R}^{n \times m}$ is a matrix $A^T \in \mathbb{R}^{m \times n}$ defined:

$$[A^T]_{ij} = A_{ij}$$

    - $(A^T)^T = A$
    - $(AB)^T = B^T A^T$
    - If $A = A^T$ then A is said to be **symmetric**.
  - **Identity**: It's a diagonal square (necessarily) matrix $I \in \mathbb{R}^n$ like this

$$\begin{pmatrix} 0 & 0 & ... & 1 \\ 0 & ... & 1 & 0 \\ ... & ... & ... & ... \\ 1 & 0 & ... & 0 \end{pmatrix}$$

- **Inverse**: The inverse of $A \in \mathbb{R}^{n \times n}$ is a matrix A^{-1} such that

$$AA^{-1} = I = A^{-1}A$$

  - A^{-1} exists only if:
    - The rank of the matrix is maximal.
    - $\det(A) \neq 0$.
  - If A is rectangular and the resulting square matrix AA^T is invertible then we have that

$$A^{\dagger} = A^T(AA^T)^{-1}$$

    is the **pseudo-inverse** matrix that satisfies $AA^{\dagger} = I$.

- **Positive Semi-definite Matrix**: A symmetric matrix $A \in \mathbb{R}^{n \times n}$ with the property that $x^T Ax \geq 0$ for any vector $x \in \mathbb{R}^n$. Its eigenvalues are all ≥ 0 .
- **Positive Definite Matrix**: A symmetric matrix $A \in \mathbb{R}^{n \times n}$ with the property that $x^T Ax > 0$ for any vector $x \in \mathbb{R}^n$ .Its eigenvalues are all > 0 .
  - **Note**: They are always invertible.

- **Linear Problems**: Given a matrix $A \in \mathbb{R}^{n \times n}$ and a vector $b \in \mathbb{R}^n$, a linear problem is the problem of finding $x \in \mathbb{R}^n$ such that:

$$Ax = b$$

We are looking for the linear combination of the columns of A giving $b = \sum_i x_i a_i$ as result. The solution is

$$x = A^{-1}b$$

- **Note**: To solve this, the time complexity is $O(n^3)$.

1.3. Optimization Problems

- **Optimization**: Maximizing or minimizing a real function $f : \mathbb{R}^n \longrightarrow \mathbb{R}$ with respect to some variables, possibly to some costraints. That is, finding the vector x such that $f(x)$ is maximal or minimal.
  - **Discrete Optimization**: variables have discrete values.
  - **Continuous Optimization**: variables have real values.

$$f(x) = g(x + \beta h(x))$$

  - **Note**: The higher β is higher the influence of $h(x)$ will be, the lower the value the higher the influence of $g(x)$.
  - **Linear Optimization Problem**:

$$\begin{aligned} \min_x \quad & f(x) = a \cdot x \\ s.t. \quad & Ax \leq b \\ & Bx = d \end{aligned}$$

  - **Quadratic Optimization Problem**:

$$\begin{aligned} \min_x \quad & f(x) = x^T Ax + bx \\ s.t. \quad & Bx \leq c \end{aligned}$$

  If the matrix A is definite positive then the function $f(x)$ is convex and the optimization problem admits a unique minimum.

2. Supervised Learning

- **Supervised Learning**: It's a type of machine learning paradigm where an algorithm learns to map input data to a specific output based on labeled data, it's called *supervised* because to test its effectiveness we give him test data of which we already know the answer but he does not.

- **Input**: $x \in X$
- **Output**: $y \in Y$
  - **Classification**:
    - **Binary**: $Y = \{-1, +1\}$
    - **Multi-class** $Y = \{1, ..., m\}$
  - **Regression**: $Y \equiv \mathbb{R}$
- **Oracle / Nature**:
  - $f : X \longrightarrow Y$ deterministic, it's ideal and **unknown**.
  - $P(x)$ or $P(y|x)$ (the stochastic version), still unknown.
- **Data**: $\text{Dataset} = \{(x_1, y_1), ..., (x_n, y_n)\}$
  - Typically **independent** and **identically distributed**.
    - **Note**: Time-series aren't independent and free of **concept drift** (the hidden relationship between X and Y is lost over time).
- **Hypothesis Space**: $H \equiv \{h \mid h : X \longrightarrow Y\}$

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